INCH-POUND

A-A-1340A
March 12, 2002
SUPERSEDING
A-A-1340
January 14, 1980

COMMERCIAL ITEM DESCRIPTION

WRENCH, BOX, STUB END TYPE, OFFSET HANDLE

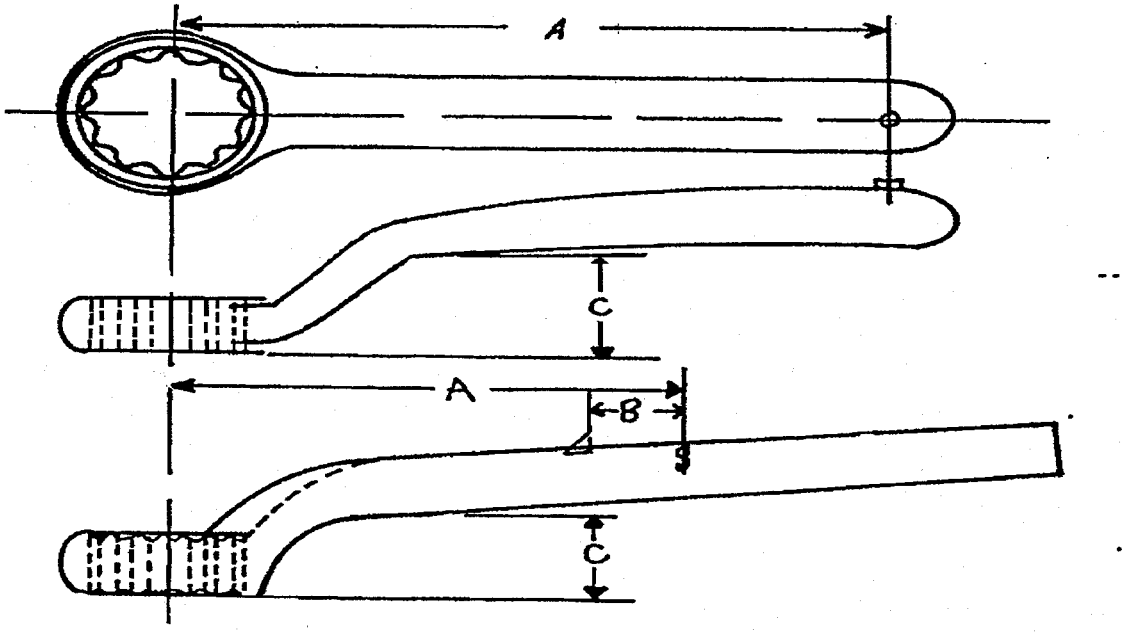
The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers wrenches suitable for heavy duty nut turning on hexagon nuts.

2. CLASSIFICATION. Not applicable

3. SALIENT CHARACTERISTICS.

3.1 Design. Wrench openings shall be of a double hexagon (12 point) design. The handle shall afford a well-proportioned, comfortable handgrip. Wrench dimensions and sizes shall conform to Section 3.2.



Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Hardware SuperStore (6FEE), 1500 E. Bannister Rd Bldg. #6, Kansas City, Missouri 64131.

FSC 5120

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3.2 Dimensions and Sizes.

Wrench Opening (Distance Across Flats)	Outside Diameter	Thickness Of Head	Wrench Opening To Plunger (Centerline To Centerline) A		Centerline Of Plunger To Stop (Stop Optional) B	Height Of Offset C	Diameter Of Handle	Overall Length	Test Load
	Max.	Max	Min	Max		Min.	-0.062	Min.	Min.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	In-Lb.
1-1/16	1-43/64	3/4	4-15/16	6-13/16	1-1/4	15/16	7/8	7-13/16	6,500
1-1/8	1-3/4	3/4	4-15/16	7-1/8	1-1/4	1	7/8	7-7/8	7,000
1-1/4	1-15/16	7/8	4-15/16	7-11/16	1-1/4	1-1/16	7/8	7-15/16	7,250
1-5/16	2	7/8	6-7/16	8	1-1/4	1-1/8	7/8	9-15/16	8,000
1-3/8	2-3/32	7/8	6-7/16	8	1-1/4	1-3/16	7/8	9-15/16	9,000
1-7/16	2-7/32	15/16	6-7/16	8-3/8	1-1/4	1-1/4	7/8	10-3/8	9,550
1-1/2	2-5/16	15/16	6-7/16	9	1-1/4	1-5/16	7/8	10-5/8	10,450
1-9/16	2-11/32	15/16	6-7/16	9-1/16	1-1/4	1-5/16	7/8	10-5/8	11,550
1-5/8	2-1/2	1	6-7/16	9-1/8	1-1/4	1-3/8	1	10-11/16	11,750
1-11/16	2-5/8	1-1/16	6-15/16	9-9/16	1-1/4	1-7/16	1	11-13/16	14,000
1-3/4	2-11/16	1-1/16	6-15/16	9-9/16	1-1/4	1-1/2	1	11-13/16	18
1-13/16	2-3/4	1-5/32	6-15/16	10-1/8	1-1/4	1-9/16	1	12-3/16	19,000
1-7/8	2-13/16	1-1/8	6-15/16	10-3/16	1-1/4	1-5/8	1	12-1/2	21,800
2	3	1-1/4	6-15/16	10-5/16	1-1/4	1-3/4	1	12-3/4	29,000
2-3/16	3-7/32	1-5/16	6-15/16	10-5/16	1-1/4	1-7/8	1	12-13/16	29,000
2-1/4	3-3/8	1-11/32	6-15/16	10-7/16	1-1/4	1-7/8	1	13	29,000
2-3/8	3-7/8	1-3/8	10-7/16	11-9/16	5-1/4	2	1-1/8	13-3/8	32,000
2-9/16	4-3/16	1-9/16	10-7/16	11-9/16	5-1/4	2-1/4	1-1/8	13-3/4	35,000
2-5/8	4-3/16	1-9/16	10-7/16	11-9/16	5-1/4	2-1/4	1-1/8	13-9/16	35,000
2-3/4	4-1/2	1-11/16	9-11/16	11-9/16	5-1/4	2-3/8	1-1/8	12-15/16	40,000
2-15/16	4-9/16	1-3/4	9-11/16	11-9/16	5-1/4	2-3/8	1-1/8	13-1/8	45,000
3-1/8	4-7/8	1-3/4	9-11/16	11-9/16	5-1/4	2-3/8	1-1/8	13-1/8	48,000

3.3 Materials. The materials used in the manufacturing of the wrenches shall be such as to produce wrenches conforming to the physical requirements hereinafter described.

3.4 Marking. Each wrench shall be marked on one of the faces or on the handle as close to each head as is practicable in a plain and permanent manner. In addition to size markings, each wrench shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.5 Hardness. Wrenches shall be treated to a Rockwell C hardness of not less that 38 nor more than 55 when tested per ASTM-E-18.

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3.6 Proof Torque. When tested as specified in ASME-B107.6, the wrenches shall withstand the proof torque specified in Section 3.2 without failure or permanent deformation that might affect the durability or serviceability of the wrench.

3.7 Wrench Opening. The nut and bolt head-engaging surfaces of the wrench shall be finished in a smooth and well-defined manner. Tolerances shall be such as to insure acceptance when gaged with gages conforming to ASME B107.17M.

3.8 Wrench Handle. Shall be of a stub end round design and suitable for accepting handle extension. The wrench handle shall be provided with a plunger stop to prevent handle extension from turning or slipping under normal working conditions. At the suppliers option, a stop may be provided.

3.9 Finish.

3.9.1 Surface Finish. All surfaces shall be thoroughly cleaned, free from cracks, and essentially free from burrs, pits, nodules, and other detrimental deficiencies.

3.9.2 Coatings. The coating shall be chrome plate (3.9.2.1) phosphate (3.9.2.2), oxide (3.9.2.3), or alternate coating (3.9.2.4) and shall be specified in the contract or order. The coating shall be adherent, smooth, continuous, and free from uncoated areas, pits, blisters, nodules, and any other defects that would interfere with their protective value and serviceability.

3.9.2.1 Chrome Plate. Wrenches shall have a protective-decorative nickel-chromium plating. The nickel thickness shall be a minimum of 0.000150 in. The chromium thickness shall be a minimum of 0.000003 in. A nickel-iron undercoating (16% iron max.) may be substituted for nickel.

3.9.2.1.1 Chrome Plate Tests.
One measurement of the thickness of each electroplated coating shall be conducted in accordance with any of the following, as appropriate, methods:

ASTM B 487: Standard Method for Measurement of metal and oxide coating thickness by microscopical examination of a cross section.

ASTM B 499: Standard Test Method for Measurement of coating thickness by the magnetic method: nonmagnetic coatings of magnetic basis metals.

ASTM B 504: Standard Method for Measurement of thickness of metallic coatings by the coulometric method.

ASTM B 530: Standard Test Method for Measurement of coating thickness by the magnetic method: electrodeposited nickel coatings on magnetic and nonmagnetic substrates.

ASTM B 568: Standard Test Method for Measurement of coating thickness by x-ray spectrometry.

ASTM B 748: Standard Method for Measurement of thickness of metallic coatings by measurement of cross section with a scanning electron microscope.

ASTM A 754: Standard Test Method for Coating thickness by x-ray fluorescence.

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Visible contact marks resulting from electroplating operations shall be confined to the interior of the wrenching and drive openings. Surfaces of the electroplated tool shall be adherent, smooth, continuous, and free of visible defects, such as blisters, pits, roughness, cracks, and uncoated areas. Surfaces shall not be stained or discolored and shall have a bright appearance.

The electroplated tool shall be clean and free of damage. The tool shall be tested for adhesion of electroplated materials in accordance with ASTM B571, Standard Test Methods for Adhesion of Metallic Coatings, by one of the following methods: Grind-Saw, Push, File, or Burnishing.

3.9.2.2 Phosphate. Wrenches shall have a chemically produced phosphate coating followed by a coating of rust preventive.

3.9.2.3 Oxide. Oxide coated wrenches shall have a coating consisting of a chemically produced oxide followed by a coating of rust preventive.

3.9.2.4 Alternative Coatings. Alternate coatings may be used in lieu of nickel - chromium and shall be subjected to the Alternative Coating Test as specified below.

3.9.2.4.1 Alternative Coating Test.

Test Preparation. The quantity and condition of the wrenches used for the following testing shall be per the manufacturer's standard practice or as mutually agreed to by the manufacturer and the General Services Administration.

Adhesion Test. Sample wrenches shall pass the file or grind-saw test of ASTM B 571, Standard practice for qualitative adhesion testing of metallic coatings.

Abrasion Test. Sample wrenches shall have no base material exposed after being subjecting to100 liters of falling sand per ASTM D 968 Method A, Standard Test Method for abrasion resistance of organic coatings by falling abrasive.

Corrosion Test. The exterior surfaces of sample wrenches shall be tested for corrosion resistance by exposure to a 72-hour salt spray test, as specified in ASTM B 117, Operating salt spray (FOG) apparatus, without falling below the ASTM B 537, Standard practice for rating of electroplated panels subjected to atmospheric exposure, rating of 6.

4. REGULATORY REQUIREMENTS.

4.1 Recovered Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The product provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawing, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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5.2 Responsibility for Inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

6. PACKAGING. Preservation, packaging and marking shall be as specified in the contract or order.

7. NOTES. This section contains information of a general or explanatory nature that is not mandatory.

7.1 Source of Documents.

American Society of Mechanical Engineers (ASME), Manuel Gutierrez, Managing Director, 3 Park Avenue, New York, NY 10016.

American Society for Testing and Materials (ASTM), 100 Barr Harbor Dr., W. Conshohocken, PA, 19428-2959.

Federal Acquisition Regulation (FAR): Government Printing Office, Superintendent of Documents, Washington DC. 20401-9371.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents.

a. Title, number, and date of Commercial Item Description.
b. Wrench Opening.
c. Type of coating
d. Packaging requirements.

7.3 National Stock Numbers (NSN's). A list of NSN's assigned that correspond to this CID. The list may not be indicative of all possible NSN's associated with the CID.

NSN	Wrench Opening	NSN	Wrench Opening
5120-00-541-1297	1-1/16	5120-00-277-7582	1-13/16
5120-00-288-9677	1-1/8	5120-00-287-4168	1-7/8
5120-00-288-9678	1-1/4	5120-00-287-4170	2
5120-00-541-1298	1-5/16	5120-00-287-4169	2-3/16
5120-00-541-1299	1-3/8	5120-00-357-8334	2-1/4
5120-00-541-1300	1-7/16	5120-00-287-4167	2-3/8
5120-00-293-0672	1-1/2	5120-00-287-4164	2-9/16
5120-00-541-1301	1-9/16	5120-00-287-4165	2-5/8
5120-00-449-8094	1-5/8	5120-00-293-0674	2-3/4
5120-00-541-1302	1-11/16	5120-00-264-5217	2-15/16
5120-00-287-4171	1-3/4	5120-00-287-4172	3-1/8

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7.4 Key words. Wrench; Box, Stub End Type, Offset Handle.

MILITARY INTERESTS:
NONE: DoD has no registered interest
in revisions and amendments to this
Commercial Item Description until further notice.

Preparing Activity:
GSA-FSS

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